AECCS: Assessing the Effectiveness of Cookie Consent Systems

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CS475 Privacy-Enhancing Technologies · Final Report · 1 March 2026

Abstract

We present AECCS, an automated measurement system that evaluates GDPR cookie consent compliance, dark pattern prevalence, and privacy-enhancing technology (PET) effectiveness across 100 popular European and international websites. Our pipeline crawls each site in three consent states (no interaction, accept all, reject all), classifies trackers using EasyList/EasyPrivacy filter lists, detects seven types of dark patterns in consent banners, and computes per-site compliance scores (0–100). We further evaluate seven browser-level PETs across all 100 sites (700 measurements), rank consent management platforms (CMPs) as privacy tools, and apply differential privacy mechanisms to protect aggregate statistics. Our findings reveal that 89.7% of sites deploy pre-consent trackers, 82% employ at least one dark pattern, and the average compliance score is just 33.1/100. Among PETs, Brave Shields provides the strongest tracker reduction (14.7%), while OneTrust-managed sites show the highest CMP effectiveness (PET score 33.4). We recommend an epsilon of 1.0 for differentially private publication of aggregate compliance statistics.

1. Introduction

Under the EU's General Data Protection Regulation (GDPR), websites must obtain explicit, informed consent before placing non-essential cookies. Users should have clear accept/reject choices, no manipulative design patterns, and no tracking activated before consent is given. In practice, enforcement is uneven and many sites bypass these requirements through dark patterns, missing reject options, or pre-consent tracker deployment.

This project addresses the question: *Do cookie consent systems on popular websites effectively protect user privacy under GDPR requirements?* We built an automated pipeline that crawls real websites from an EU IP address, measures compliance signals, evaluates PET countermeasures, and produces actionable findings.

2. Related Work

AECCS builds on three strands of prior work. Nouwens et al. [1] found that only 11.8% of consent banners met minimal GDPR requirements, with dark patterns actively steering users toward acceptance. Matte et al. [2] demonstrated that many CMPs store positive consent before users interact with the banner. Bollinger et al. [3] developed CookieBlock, an automated system for classifying and blocking non-essential cookies. Our work extends these by combining compliance measurement, dark pattern detection, and PET evaluation into a single automated pipeline, and by applying differential privacy to aggregate reporting.

3. Methodology

3.1 Website Selection

We compiled a list of 100 websites across 11 categories: news (20), e-commerce (15), social media (10), entertainment (9), education (9), travel (8), tech (8), finance (6), government (5), marketplace (5), and employment (2). Sites were selected from European and international rankings, focusing on EU-targeted domains.

3.2 Data Collection

We used Playwright for headless browser automation with Proton VPN (Netherlands) for EU geolocation. Each site was visited in three states: (1) no interaction with the consent banner, (2) clicking "Accept All," and (3) clicking "Reject All." For each state we recorded all cookies, HTTP requests, third-party domains, and consent banner HTML. Anti-bot measures were mitigated with 8 rotating user agents, randomized 2–5 second delays, and stealth mode.

3.3 Analysis Pipeline

The pipeline consists of 11 automated stages: crawl, classify, dark pattern detection, scoring, metrics, PET evaluation, differential privacy reporting, CMP analysis, unified comparison, visualization (11 figures), and HTML report generation. All stages are orchestrated by a unified pipeline runner with manifest-based tracking for reproducibility.

Module	Function	Status
Crawler	Playwright-based 3-state crawling	Completed
Classifier	EasyList/EasyPrivacy/Disconnect matching	Completed
Dark Patterns	7 pattern types via HTML analysis	Completed
Scoring	6-criteria weighted compliance (0–100)	Completed
Metrics	Aggregate statistics computation	Completed
Browser PETs	7 PET configs × 100 sites	Completed
DP Reporting	Laplace/Gaussian/Randomized Response	Completed
CMP Analysis	CMP effectiveness ranking	Completed
Visualization	11 publication-quality figures	Completed

3.4 Compliance Scoring

Each site receives a score from 0 to 100 based on six weighted criteria: no pre-consent trackers (25%), reject option available (20%), equal accept/reject effort (15%), no dark patterns (15%), post-reject compliance (15%), and transparent information (10%). Letter grades are assigned: A (90+), B (75+), C (60+), D (40+), F (<40).

4. Results

4.1 Pre-Consent Tracker Violations

Of 97 successfully crawled sites, 87 (89.7%) deployed trackers before any user interaction with the consent banner. The average site loaded 6.7 pre-consent trackers (median: 5). This represents a clear violation of GDPR's requirement for prior consent.

Violation rates varied by category. Entertainment, finance, travel, and employment sites showed 100% violation rates. News sites averaged the highest tracker density (11.7 per site), followed by education (10.9) and travel (8.4). Government sites had the lowest average (1.2 trackers), though 80% still deployed at least one.

4.2 Tracker Ecosystem

We identified 403 unique tracker cookies across 93 tracker domains. By category, 54.3% were analytics trackers, 26.3% advertising, 14.6% social, 4.2% fingerprinting, and 0.5% functional. The dominant vendors were:

Vendor	Sites present	Share
Google	31	32.0%
Meta	10	10.3%
Adobe	6	6.2%
Amazon	6	6.2%
LinkedIn	6	6.2%
Criteo	4	4.1%
Hotjar / X / TikTok	3 each	3.1% each

4.3 Consent Banner and Reject Effectiveness

72 of 97 sites displayed a consent banner. Of these, only 31 (43.1%) provided a visible "Reject All" button. 28 sites (38.9%) offered equal click effort for accept and reject. Critically, clicking "Reject All" reduced tracker counts on only 7.2% of sites and eliminated all trackers on just 9.3%. On average, sites retained 7.0 trackers after rejection compared to 6.7 before interaction—meaning rejection was largely ineffective.

4.4 Dark Patterns

82 sites (82.0%) employed at least one dark pattern in their consent interface. The breakdown by type:

Dark Pattern	Sites	Prevalence
Missing reject option	69	69.0%
Multi-layer rejection	14	14.0%
Asymmetric buttons	7	7.0%
Hidden reject	6	6.0%
Confusing language	5	5.0%
Forced action	1	1.0%

The dominant dark pattern is the complete absence of a reject button (69%), forcing users to navigate through settings menus to decline. Multi-layer rejection, where rejecting requires more clicks than accepting, affected 14% of sites.

4.5 Compliance Scores

The average compliance score was 33.1/100 (median: 24.0, std dev: 21.7). The score range was 9.0 to 75.0—notably, no site achieved an A grade (≥90).

Grade	Score range	Count	Percentage
A	90–100	0	0.0%
B	75–89	1	1.0%
C	60–74	18	18.6%
D	40–59	13	13.4%
F	0–39	65	67.0%

Two-thirds of sites received a failing grade. By category, finance (47.8) and entertainment (47.0) scored highest, while marketplace (20.4) and news (23.3) scored lowest:

Category	Avg Score	Median
Finance	47.8	60.0
Entertainment	47.0	60.5
Technology	44.6	46.2
Government	35.9	24.5
Education	35.7	34.0
Employment	35.5	35.5
E-Commerce	32.9	24.0
Social Media	32.4	29.5
Travel	23.4	14.0
News	23.3	14.8
Marketplace	20.4	19.0

4.6 CMP Distribution

We identified six commercial CMP providers across the 100 sites:

CMP	Sites
Custom/Unknown	43
OneTrust	26
TrustArc	17
Cookiebot	6
Didomi	3
Quantcast	1
Usercentrics	1

43% of sites use custom or unidentifiable consent implementations rather than established CMP providers, which correlates with lower compliance scores and fewer reject options.

5. PET Evaluation

5.1 Browser PET Comparison

We tested 7 PET configurations across all 100 sites (700 total measurements, 681 successful). Each configuration was evaluated by measuring tracker cookies, tracker domains, and third-party requests compared to the baseline (no PET).

Rank	PET	Avg Tracker Domains	Avg Tracker Cookies	Reduction %
1	Brave Shields	2.0	0.1	14.7%
2	Firefox ETP Strict	3.4	1.4	−8.7%
3	uBlock Origin	3.3	1.2	−13.7%
4	Firefox ETP Standard	3.4	1.4	−16.3%
5	Consent-O-Matic	3.4	1.3	−17.7%
6	Privacy Badger	3.4	1.3	−18.4%

Brave Shields was the only PET that achieved a positive average tracker reduction, primarily through aggressive route-level blocking of known tracker domains. The negative reduction percentages for other PETs indicate that these tools did not substantially reduce tracker exposure relative to baseline in this measurement configuration.

5.2 CMP Effectiveness as Privacy Tools

We ranked CMPs as privacy tools using a composite PET score (30% compliance score, 25% reject effectiveness, 20% reject availability, 15% inverse dark patterns, 10% tracker reduction):

CMP	PET Score	Compliance	Reject Available
OneTrust	33.4	45.1	61.5%
TrustArc	24.1	37.2	47.1%
No CMP Detected	14.9	28.3	15.2%
Cookiebot	5.6	18.8	0.0%
Usercentrics	5.0	16.5	0.0%
Didomi	4.0	14.0	0.0%
Quantcast	2.2	18.5	0.0%

OneTrust-managed sites showed the highest compliance and the most frequent presence of a reject button (61.5%). Cookiebot, Usercentrics, Didomi, and Quantcast sites in our sample had 0% reject button availability, resulting in very low PET scores.

5.3 Best PET+CMP Combinations

The most effective combination was Brave Shields on a site using OneTrust, with an estimated 43.2% combined effectiveness. The top 5 combinations:

Browser PET	CMP	Est. Effectiveness
Brave Shields	OneTrust	43.2%
Brave Shields	TrustArc	35.3%
Firefox ETP Strict	OneTrust	27.6%
Brave Shields	No CMP	27.4%
uBlock Origin	OneTrust	24.3%

6. Differential Privacy Analysis

To enable privacy-preserving publication of aggregate compliance statistics, we applied three DP mechanisms: Laplace (for numeric queries), Gaussian (with delta parameter), and Randomized Response (for binary attributes). We tested epsilon values from 0.1 to 10.0 over 100 trials each across 38 aggregate queries.

Epsilon	Avg MAE	Utility
0.1	25.10	Low (high privacy)
0.5	12.25	Moderate
1.0	7.23	Balanced (recommended)
2.0	3.95	Good utility
5.0	1.76	High utility
10.0	0.86	Near-exact

We recommend epsilon = 1.0 as the best balance between privacy protection and statistical utility. At this level, the average mean absolute error is 7.23 across all queries, which preserves the key compliance trends while providing meaningful privacy guarantees for individual site data.

7. Discussion

Our findings paint a concerning picture of GDPR cookie consent compliance. Nearly 9 in 10 sites deploy trackers before any consent interaction, and two-thirds receive a failing compliance grade. The most common dark pattern—simply omitting a reject button—affects 69% of sites, making meaningful rejection impossible for most users.

The reject mechanism itself is largely theatrical: clicking "Reject All" reduces trackers on only 7.2% of sites. This aligns with Matte et al.'s findings that CMPs often fail to enforce rejection technically, even when the option exists.

Among PETs, Brave Shields was the only tool that achieved meaningful tracker reduction in our measurement setup. This suggests that users cannot rely on consent banners alone for privacy protection and should use browser-level blocking tools as a complement.

The CMP landscape shows that OneTrust provides the most compliant consent interfaces, while smaller CMPs in our sample failed to provide basic reject functionality. Sites without any identified CMP performed worst overall.

8. Limitations and Future Work

- Our sample of 100 sites, while covering 11 categories, may not fully represent the broader web. Future work should scale to 500+ sites.
- The crawl captures a single snapshot in time; consent systems may change behavior based on user history or time of day.
- Some PET measurements showed negative "reduction" values, likely because extensions alter page load timing and can trigger different tracker behavior. Longer observation windows could mitigate this.
- Dark pattern detection relies on HTML/CSS analysis and keyword matching. More sophisticated visual analysis (e.g., screenshot-based ML classifiers) could improve accuracy.
- The study was conducted from a single EU location (Netherlands). Multi-country measurements could reveal geolocation-dependent consent behavior.

9. Conclusion

AECCS demonstrates that cookie consent systems on popular websites largely fail to protect user privacy under GDPR. With 89.7% pre-consent violation rates, 82% dark pattern prevalence, and an average compliance score of just 33.1/100, the current consent ecosystem provides more of an illusion of control than actual protection. Browser-level PETs like Brave Shields offer some mitigation, and differential privacy enables responsible publication of these aggregate findings. Our automated pipeline is fully reproducible and can be re-run to track compliance trends over time.

References

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